

## MODULE 1 TISSUES

# Histology: The Four Tissue Types.

## TISSUES &amp; HISTOLOGY

Tissues are groups of similar cells performing a specific function. Four types build every organ in the body, and the arrangement of the cells is what predicts what the tissue can do.

**COMPETENCIES** W1-GERM-LAYERS, W1-EPITHELIAL-ID, W1-CELL-JUNCTIONS, W1-CONNECTIVE-ID, W1-BODY-MEMBRANES

**BY THE END**

1. Name the four tissue types, their general functions, and where each is found.
2. Define a biopsy and walk through the procedure.
3. Identify the five cell junctions, their proteins, locations, and functions.
4. Separate epithelial from connective tissue on structure alone.
5. Classify epithelium by layers and cell shape, and name a location for each type.
6. Classify glands structurally and functionally.
7. Name the cells, fibers, and ground substance of connective tissue and what each contributes.
8. Recognize cartilage, bone, muscle, and nervous tissue at survey level.



WATCH AFTER YOU READ THIS CHAPTER

## Concept videos: Tissues

[drsrennie-stack.github.io/new-build-bio4-solano/tissues-concept-videos.html](https://drsrennie-stack.github.io/new-build-bio4-solano/tissues-concept-videos.html)

## The Four Major Tissue Types

1. **Epithelial tissue**, covers body surfaces, lines cavities, and forms glands.
2. **Connective tissue**, protects, supports, and binds structures and organs.
3. **Nervous tissue**, detects changes and generates nerve impulses.
4. **Muscle tissue**, facilitates movement and generates force.

### Cells that stay and cells that move

Most cells are **stationary**. Some **migrate**, particularly during growth and development. White blood cells migrating to a site of injury or infection are the everyday example.

## Biopsy

**Definition.** Removal of a tissue sample for microscopic examination. This is how tissue-level questions get answered clinically.

1. Collect a tissue sample using a needle, scalpel, or endoscope.
2. Preserve the tissue in a fixative such as formalin.
3. Section the tissue into thin slices for examination.
4. Stain the tissue for contrast under a microscope.

## The Five Cell Junctions

| JUNCTION, PROTEIN, LOCATION, FUNCTION |                                                                                                                           |                                                                                                               |                                                                                 |                        |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------|
| Junction                              | What it is                                                                                                                | Function                                                                                                      | Location                                                                        | Major protein          |
| <b>Tight junction</b>                 | A network of transmembrane proteins that fuse together on adjacent plasma membranes, sealing the passageway between cells | Prevents substances passing between cells, prevents organ contents leaking into surrounding tissue            | Epithelium lining the stomach, intestines, urinary bladder, blood-brain barrier | Claudins and occludins |
| <b>Adherens junction</b>              | Adhesion belts formed by transmembrane proteins attached to microfilaments                                                | Provides mechanical stability, helps epithelial surfaces resist separation during contractile activity        | Intestinal epithelial cells                                                     | Cadherins              |
| <b>Desmosome</b>                      | Plaques and cadherins connected to intermediate filaments                                                                 | Strongly binds cells to each other, prevents separation under tension                                         | Outer epidermis, cardiac muscle cells                                           | Cadherins              |
| <b>Hemidesmosome</b>                  | Anchors cells to the basement membrane                                                                                    | Prevents separation of epidermis from basement membrane, stabilizes epithelial cells during mechanical stress | Skin, heart muscle                                                              | Integrins              |
| <b>Gap junction</b>                   | Protein channels connecting neighboring cells, allowing passage of small ions and molecules                               | Cell to cell communication, transfers nutrients and waste, allows impulses to spread                          | Lens, cornea, embryonic tissues, neurons, cardiac tissue, GI tract, uterus      | Connexins              |

### HOLD ONTO THIS

Desmosomes connect cell to cell. Hemidesmosomes connect cell to basement membrane. Half the name, half the job.

## Epithelium versus Connective Tissue

| THE TWO TISSUE TYPES COMPARED |                                                    |                                                          |
|-------------------------------|----------------------------------------------------|----------------------------------------------------------|
| Feature                       | Epithelial tissue                                  | Connective tissue                                        |
| <b>Matrix and cells</b>       | Tightly packed cells, minimal extracellular matrix | Large extracellular matrix, few cells                    |
| <b>Blood vessels</b>          | Avascular, relies on diffusion                     | Highly vascular, with cartilage and tendon as exceptions |
| <b>Location</b>               | Covers surfaces and forms linings                  | Supports and connects other tissues                      |

**Similarities.** They work together for nutrient exchange, since avascular epithelium depends on the vascular connective tissue beneath it. Both play critical roles in maintaining structure and function.

## Epithelial Tissue

### Three functions of epithelium

- **Selective barrier**, controls transfer of substances into and out of the body.
- **Secretory surface**, produces and releases cellular products.
- **Protective surface**, provides resistance against abrasion and environmental influences.

### The basement membrane

**Location.** Below the basal surface of epithelial tissue. **Composition.** A **basal lamina** of collagen, laminin, and glycoproteins, plus a **reticular lamina** of collagen and glycoproteins.

**Functions.** Supports and anchors epithelial tissue, facilitates wound healing and cell migration, restricts large molecules, and filters substances in the kidneys.

### Classification of epithelium

| EPITHELIAL TYPES, FUNCTION, AND LOCATION  |                                                                                               |                                                  |                                                     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------|
| Tissue type                               | Description                                                                                   | Function                                         | Locations                                           |
| <b>Simple squamous</b>                    | Single layer of flat, thin cells                                                              | Filtration, diffusion, osmosis, secretion        | Alveoli, glomeruli, lining of blood vessels, serosa |
| <b>Simple cuboidal</b>                    | Single layer of cube-shaped cells                                                             | Secretion and absorption                         | Kidney tubules, small glands, ovary surface         |
| <b>Simple columnar</b>                    | Single layer of tall, column-shaped cells, may have microvilli or goblet cells                | Absorption, secretion of mucus and enzymes       | Digestive tract from stomach to rectum, gallbladder |
| <b>Stratified squamous</b>                | Several layers, apical layers are flat                                                        | Protection against abrasion                      | Skin (keratinized), esophagus, mouth, vagina        |
| <b>Stratified cuboidal</b>                | Two or more layers of cube-shaped cells                                                       | Protection and limited secretion                 | Sweat glands, mammary glands, salivary glands       |
| <b>Stratified columnar</b>                | Several layers, apical cells are columnar                                                     | Protection and secretion                         | Male urethra, ducts of some glands                  |
| <b>Transitional</b>                       | Several layers, cells stretch and change shape from dome-shaped to flat                       | Stretching to allow distension                   | Urinary bladder, ureters, part of the urethra       |
| <b>Pseudostratified ciliated columnar</b> | Appears stratified but all cells touch the basement membrane, may have cilia and goblet cells | Secretion of mucus, propulsion by ciliary action | Respiratory tract (trachea, bronchi), sperm ducts   |

**HOLD ONTO THIS**

Name the layers first, then the shape of the apical cells. Simple means one layer, stratified means more than one, and the shape you name is always the shape at the free surface.

## Glandular Epithelium

### Structural classification

#### GLANDS BY NUMBER OF CELLS

| Type          | Description                        | Example                           |
|---------------|------------------------------------|-----------------------------------|
| Unicellular   | Single-celled glands               | Goblet cells, which secrete mucus |
| Multicellular | Multiple cells forming a structure | Salivary glands, sweat glands     |

**Duct type.** Simple means no branching. Compound means branching. **Shape of the secretory portion.** Tubular means tube-shaped. Acinar means rounded. Combine the two and you have the full structural name.

### Endocrine and exocrine

**Endocrine glands** release their product into the bloodstream, with no duct. Thyroid, adrenal, pituitary.

**Exocrine glands** secrete their products into ducts that empty onto a surface or into a lumen. Sweat, salivary, sebaceous. Some glands are **mixed**, doing both, and the pancreas is the classic example: exocrine digestive enzymes into a duct, endocrine insulin and glucagon into the blood.

### Functional classification of exocrine glands

#### HOW THE PRODUCT ACTUALLY LEAVES THE CELL

| Type      | Mechanism                                                                                                           | Example                   |
|-----------|---------------------------------------------------------------------------------------------------------------------|---------------------------|
| Merocrine | Secretions released via exocytosis. The cell is unharmed. Most common type                                          | Salivary glands, pancreas |
| Apocrine  | Secretions accumulate at the apical surface, which pinches off to release the product. The cell then repairs itself | Mammary glands            |
| Holocrine | Secretions accumulate in the cytoplasm and the entire cell ruptures to release the product                          | Sebaceous glands          |

**MEMORY TOOL**

HOL equals WHOLE cell is secreted.

## Connective Tissue

### Functions

- Binds together, supports, and strengthens body tissues.
- Protects and insulates internal organs.

- Compartmentalizes structures, as in skeletal muscle.
- Serves as a major transport system. Blood is a connective tissue.
- Is the primary location of stored energy reserves, as in adipose tissue.
- Is a main source of immune responses.

## The two basic elements

Connective tissue is **extracellular matrix** plus **cells**. The matrix divides into **ground substance** and **protein fibers**. The proportions of these elements are what make blood liquid, cartilage firm, and bone hard.

## Ground substance

**Definition.** The material between cells and protein fibers. **Structure.** Fluid, semifluid, gelatinous, or calcified.

**Composition.** Water and organic molecules such as chondroitin sulfate and glucosamine.

**Functions.** Supports and binds cells, supplies water and nutrients for exchange between blood and cells, and plays a role in tissue development and metabolism.

**The matrix determines the quality.** Cartilage is firm but pliable. Bone is hard and inflexible. Same basic plan, different matrix.

## The three fiber types

| COLLAGEN, RETICULAR, AND ELASTIC FIBERS |                                                                                   |                                                                             |                                                                                                                                                                                                    |
|-----------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fiber                                   | Protein                                                                           | Properties                                                                  | Locations and functions                                                                                                                                                                            |
| <b>Collagen</b>                         | Collagen, the most abundant protein in the body at about 25 percent               | Very strong, resists tension. Parallel bundles provide tensile strength     | Most connective tissues: bone, tendon, cartilage                                                                                                                                                   |
| <b>Reticular</b>                        | Collagen covered with glycoproteins                                               | Fine, branching networks                                                    | Walls of blood vessels, surrounding cells in areolar, adipose, nervous, and smooth muscle tissue. Provides support and strength, forms the stroma of soft organs, helps form the basement membrane |
| <b>Elastic</b>                          | Elastin, with the glycoprotein fibrillin providing tensile strength and stability | Stretches up to 150 percent of relaxed length and returns to original shape | Skin, lungs, blood vessel walls                                                                                                                                                                    |

## Cells of connective tissue

| WHO LIVES IN THE MATRIX  |                                                       |                                                                                                                                                                                     |
|--------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cell                     | Origin or definition                                  | Function and location                                                                                                                                                               |
| <b>Fibroblasts</b>       | Found in all general connective tissues               | Migrate through connective tissue secreting matrix fibers and ground substance                                                                                                      |
| <b>Macrophages</b>       | Irregular white blood cells derived from monocytes    | Engulf bacteria and debris via phagocytosis. Fixed macrophages stay in one tissue such as liver, lungs, spleen. Wandering macrophages migrate to sites of infection or inflammation |
| <b>Mast cells</b>        | Found along blood vessels supplying connective tissue | Secrete histamine, which dilates blood vessels during inflammation. Also bind, ingest, and kill bacteria                                                                            |
| <b>Plasma cells</b>      | Develop from B lymphocytes                            | Secrete antibodies. Found in connective tissue of the GI tract, respiratory tract, and lymphatic tissues                                                                            |
| <b>Adipocytes</b>        | Connective tissue cells that store triglycerides      | Found deep in the skin and around organs such as the heart and kidneys                                                                                                              |
| <b>White blood cells</b> | Various leukocytes                                    | Immune defense                                                                                                                                                                      |

## Blasts versus cytes

| IMMATURE AND MATURE FORMS |                                                  |                                          |
|---------------------------|--------------------------------------------------|------------------------------------------|
|                           | Immature, the blasts                             | Mature, the cytes                        |
| <b>Examples</b>           | Fibroblasts, osteoblasts, chondroblasts          | Fibrocytes, osteocytes, chondrocytes     |
| <b>What they do</b>       | Actively divide and secrete extracellular matrix | Maintain and repair extracellular matrix |

## General features of connective tissue

- **Location.** Found below epithelial tissue, never on body surfaces.
- **Vascularity.** Highly vascular. Exceptions: cartilage and tendon.
- **Innervation.** Supplied with nerves. Exceptions: cartilage and blood.

## Embryonic mesenchyme and mucous connective tissue

| THE TWO EMBRYONIC CONNECTIVE TISSUES |                                                                      |                                                     |
|--------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------|
|                                      | Embryonic mesenchyme                                                 | Mucous connective tissue                            |
| <b>Location</b>                      | In the embryo, under skin and along bones                            | In the umbilical cord                               |
| <b>Significance</b>                  | Differentiates into all types of connective tissue                   | Provides support                                    |
| <b>Characteristics</b>               | Irregular cells in a semifluid matrix with delicate reticular fibers | Widely scattered fibroblasts in a jelly-like matrix |

## Blood, a Fluid Connective Tissue

### FORMED ELEMENTS AND WHAT EACH DOES

| Cell type                 | Function                                  |
|---------------------------|-------------------------------------------|
| Neutrophils               | Phagocytosis of bacteria                  |
| Eosinophils               | Combat parasitic infections and allergies |
| Lymphocytes               | Fight viruses                             |
| Monocytes and macrophages | Phagocytosis, transform into macrophages  |
| Basophils                 | Involved in allergic responses            |
| Red blood cells           | Transport oxygen and carbon dioxide       |
| Platelets                 | Facilitate blood clotting                 |

## Connective Tissues, Cartilage, Bone, and Nervous Tissue

### SURVEY OF THE REMAINING TISSUE TYPES

| Tissue type               | Description                                                                                         | Function                                                   | Locations                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|
| <b>Areolar connective</b> | Loose arrangement of collagen, elastic, and reticular fibers with scattered cells                   | Strength, elasticity, support                              | Under epithelial tissue, around organs and blood vessels    |
| <b>Adipose connective</b> | Cells filled with fat droplets, sparse extracellular matrix                                         | Energy storage, insulation, cushioning                     | Subcutaneous tissue, around kidneys and heart               |
| <b>Hyaline cartilage</b>  | Glassy appearance, fine collagen fibers not visible with ordinary staining, chondrocytes in lacunae | Smooth surface for joint movement, flexible support        | Ends of long bones, costal cartilage, nose, trachea, larynx |
| <b>Fibrocartilage</b>     | Thick collagen fibers dominate                                                                      | Resists compression, absorbs shock                         | Intervertebral discs, pubic symphysis, knee menisci         |
| <b>Elastic cartilage</b>  | Similar to hyaline but with more elastic fibers                                                     | Maintains shape and provides flexibility                   | External ear (pinna), epiglottis                            |
| <b>Compact bone</b>       | Hard, calcified matrix with collagen fibers, osteocytes in lacunae                                  | Supports, protects, stores calcium, houses blood formation | Bones                                                       |
| <b>Nervous tissue</b>     | Neurons and glial cells, branching neurons with long processes                                      | Transmission of electrical signals                         | Brain, spinal cord, nerves                                  |

## Muscle Tissue

### THE THREE MUSCLE TISSUE TYPES

| Tissue type            | Description                                                             | Function                                 | Locations                                             |
|------------------------|-------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------|
| <b>Skeletal muscle</b> | Long, cylindrical, multinucleated cells, striated                       | Voluntary movement, heat production      | Attached to bones and occasionally skin               |
| <b>Cardiac muscle</b>  | Branching cells connected by intercalated discs, striated, uninucleated | Involuntary contraction to pump blood    | Heart walls                                           |
| <b>Smooth muscle</b>   | Spindle-shaped cells, non-striated, uninucleated                        | Involuntary movement, propels substances | Walls of hollow organs such as intestines and bladder |

### TISSUE STRUCTURE PREDICTS HOW IT HEALS AND HOW IT FAILS

Vascularity sets healing speed. Cartilage and tendon are avascular or nearly so, which is why a torn meniscus or a ruptured Achilles heals slowly while a skin wound closes in days. Junction type sets what tears. Loss of desmosomal adhesion in the epidermis produces blistering, because the cells separate from each other. Loss of hemidesmosomes separates the whole epidermis from the basement membrane, which is a deeper and more serious blister. Cell arrangement sets the barrier. Stratified squamous epithelium survives abrasion because it sheds its top layer. Simple squamous cannot, which is why the alveoli, one cell thick for gas exchange, are so easily damaged.